Lecture 6: How Do We Communicate?: Language in the Brain, Mouth and the Hands

Introduction to Psychology · Yale University, Spring 2007 · Professor Paul Bloom

Study guide based on the lecture transcript. Source: https://oyc.yale.edu/psychology/psyc-110/lecture-6 (CC BY-NC-SA 3.0).

Before you start

This lecture is about human language: what all languages share, how they are built, and how children learn them. Professor Bloom says that language has been a "battleground" for theories of human nature. Any theory of what people are has to explain language.

Learning goals

After this lecture you should be able to:

- Explain why many scientists think language is part of human nature.
- Describe the three levels of language: phonology, morphology and syntax.
- Explain how recursion lets us make an unlimited number of sentences.
- Describe Chomsky's claim that language "grows," and the evidence for it.
- Describe the main steps of language development, from newborns to puberty.

Key ideas

1. The Scientific Notion of Language and Structure

- Bloom uses "language" in a narrow, scientific sense: systems like English, Dutch, Navajo and American Sign Language. If the word covers everything, even traffic signals, it is useless for science.
- All languages share deep universals (for example, all can express abstract ideas), but they also sound very different. A good theory must explain both facts.
- Darwin wrote: "Man has an instinctive tendency to speak… while no child has an instinctive tendency to bake, brew or write."
- Evidence: every human society has language, and every neurologically normal person has it. When slaves with different languages were mixed, adults made a **pidgin**, but their children created a full language, a **creole**, in one generation. Children in Nicaragua did the same with sign language. Brain damage can cause aphasias (loss of language), and mutations in certain genes can stop people from learning language.
- Language is creative. We understand sentences we have never heard before, so we must have unconscious rules, not a list. Compare "The pig is eager to eat" and "The pig is easy to eat."

2. Phonology: A System of Sounds

- Phonology is the system of sounds (or signs). English has about forty phonemes (basic speech sounds). It separates "l" and "r" ("lip" and "rip"); other languages do not.
- The pauses we hear between words are a psychological illusion; an oscilloscope (a machine that measures sound vibrations) shows none. A language you do not know sounds like one long stream.
- Children make mistakes with word boundaries, for example "I'll never be your pizza burnin'" for "beast of burden."
- **Top-down processing:** what you know changes what you hear. Many people hear "Newsweek magazine" in Rick James's "Super Freak," but the notes say "new wave magazines." We also hear "sentence" even when a cough covers part of it.

3. Morphology: A System of Words

- Ferdinand de Saussure called it "the arbitrariness of the sign": any sound or sign can be linked to any idea. A word for "country" cannot sound like a country.
- A morpheme is the smallest unit of meaning. "Dogs" and "complained" are one word but two morphemes.
- The average speaker knows at least 60,000 words (Bloom thinks closer to 80,000 or 100,000), about nine new words a day from age one.

4. Syntax: Communicating Complicated Ideas

- Syntax is the set of rules that combine words into phrases and sentences. Wilhelm von Humboldt called it "the infinite use of finite media."
- It is a combinatorial system (a few parts combined in many ways, like notes in music) that uses **recursion**. A toy language with three nouns (Fred, Barney, Wilma), two verbs and one rule makes only 18 sentences. Add a rule that puts a sentence inside a sentence ("Fred thinks Barney likes Wilma"), and the number becomes infinite. There is no longest sentence.
- Sentences can be ambiguous: "I once shot an elephant in my pajamas" (Groucho Marx), "Kids make nutritious snacks."
- In a murder case, a man shouted "Let him have it" while his brother pointed a gun at a police officer. It could mean "shoot him" or "give him the gun."

5. Question and Answer on Language Structure

Syntax means the same as grammar. Children with no one to talk to do not develop language, but Susan Goldin-Meadow found that deaf children with deaf siblings create their own language. Grammars have limits (no language has a rule that "the fifth word has to be a verb"), but they still allow infinite sentences.

6. Noam Chomsky and Language Acquisition

- Chomsky claims language is not really learned. It grows, like arms or a visual system.
- But experience matters: you must hear English to know English. Languages differ in sounds, words ("dog," *chien*) and word order. English is subject-verb-object ("Bill hit John"); more languages are subject-object-verb ("Bill John hit").

- In **specific language impairment**, a genetic disorder, children who are smart, social and want to communicate still cannot learn language normally. So language is partly separate from general intelligence.
- Children need no training. In some communities adults do not talk to babies, and the children still learn. Brown and Hanlon (1970s) found that parents react to the message or cuteness of what children say, not to grammar ("I loves you, Mommy").

7. The Time Course of Language Acquisition

- Four-day-old babies in France sucked a pacifier harder to hear French than Russian; in Russia, babies preferred Russian. They respond to rhythm, not words.
- Young babies hear differences between the phonemes of all languages. By about twelve months, they keep only those of their own language.
- About seven months: babbling. Sign languages and spoken languages have the same milestones.
- About twelve months: first words ("dog," "up," "milk"). About eighteen months: faster word learning and mini-sentences ("Want cookie").
- From about age seven to puberty, the ability declines. Elissa Newport and Sam Supalla found that immigrants' English depends mainly on the age they started, not on intelligence or motivation.

Key vocabulary

Term	Meaning in simple English	Example from the lecture
universal	a feature that all languages share	every language can express abstract ideas
pidgin	a simple mix of words from different languages; not a full language	used by slaves from different language backgrounds
creole	a full language that children create from a pidgin	made in a single generation
aphasia	loss of language because of brain damage	after a stroke or a blow to the head
phoneme	a basic sound that can change one word into another	"l" and "r" in "lip" and "rip"
top-down processing	using what you already know to interpret what you hear or see	hearing "Newsweek magazine" in a song
morpheme	the smallest unit of meaning	"dogs" = "dog" + plural "s"
arbitrary	with no natural link; a historical accident	"dog" in English, *chien* in French
syntax	the rules that combine words into sentences (grammar)	"Bill hit John" vs. "Bill John hit"
recursion	a rule that can use itself again and again	"Fred thinks Barney likes Wilma"
ambiguous	having more than one possible meaning	"Let him have it"

Term	Meaning in simple English	Example from the lecture
specific language impairment	a genetic disorder that makes it very hard to learn language	smart, social children who cannot learn to speak normally

Check your understanding

1. What did Darwin say about speaking compared with baking, brewing or writing?
2. What happens to children who grow up hearing only a pidgin? Why is this important?
3. Name the three levels of language structure and give one example of each.
4. Why can't our knowledge of language be just a list of sentences?
5. Use the Fred, Barney and Wilma example to explain recursion.
6. What does specific language impairment tell us about language and intelligence?
7. Compare a young baby and a twelve-month-old. Which speech sounds can each one tell apart?
8. A five-year-old and a twenty-five-year-old move to a new country. Based on Newport and Supalla's work, who will probably speak the new language better after thirty years? Why?

Answer guide

1. Humans have an instinctive tendency to speak, but no child has an instinctive tendency to bake, brew or write.
2. They create a full language, a creole, in one generation. This suggests language is part of human nature and does not need a long cultural history.
3. Phonology (sounds, like "l" and "r"), morphology (words and morphemes, like "dog" + "s"), and syntax (rules for sentences, like subject-verb-object order).
4. Because we produce and understand sentences that nobody has said before. We need unconscious rules that combine words.
5. With one simple rule, three nouns and two verbs give only 18 sentences. A rule that puts a sentence inside another sentence can repeat forever, so the number becomes infinite.
6. Some children are smart, social and want to communicate, but still cannot learn language. So language is partly separate from intelligence.
7. The young baby can tell apart the phonemes of all languages. By twelve months, the baby is sensitive mainly to the sounds of its own language.
8. The five-year-old. The main factor is the age when you start; after puberty it is rare to speak a new language without an accent.

English practice

Useful phrases from the lecture

- "It turns out..." - introduces a surprising research result. *It turns out parents respond not to the grammatical correctness...*
- "What this means is..." - explains an idea in simpler words.
- "Well, should we believe this?" - a question that introduces the evidence for a claim.
- "On the other hand, ..." - introduces the other side of an argument.
- "This suggests that..." - draws a careful conclusion from evidence.

Speaking task (with your AI tutor)

- Explain in 4-6 sentences why Darwin thought language is an instinct. Use: *pidgin*, *creole*, *human nature*.
- Explain in 4-6 sentences how finite words can make infinite sentences. Use: *rule*, *recursion*, *finite*.

Writing task

- Do parents need to teach their children language? Write one paragraph (80-120 words). Use two pieces of evidence from the lecture, for example communities that do not talk to babies and the study by Brown and Hanlon.

Study with your tutor

Things to say to your tutor in Live conversation:

- "Let's review Lecture 6. Ask me questions about phonology, morphology and syntax, one at a time."
- "Quiz me on the key vocabulary of Lecture 6."
- "I will explain recursion in my own words. Check my understanding and my English."
- "Ask me about the stages of language development, from newborn babies to puberty."

Readings from the original course

- Gray, Peter. Psychology (5th edition), chapter 11
- Pinker, Steven. "The Language Instinct: How the Mind Creates Language." In The Norton Psychology Reader. Edited by Gary Marcus. New York: W. W. Norton & Company, 2006. pp. 97-107
- Baker, Mark C. "The Atoms of Language: The Mind's Hidden Rules of Grammar." In The Norton Psychology Reader. Edited by Gary Marcus. New York: W. W. Norton & Company, 2006. pp. 108-114

My notes